

Complex numbers — arithmetic, conjugates and the Argand diagram [Cambridge revision]

One new symbol, $i^2 = -1$, and every quadratic finally has two roots.

LESSON 92–94

3 × 40 MIN

ALGEBRA 11, QO‘SHIMCHA BO‘LIM

P2/P3 11.1–11.4

LESSON CLOCK

20'

30'

30'

30'

25'

5

The new number	20 min
Arithmetic	30 min
The conjugate and division	30 min
The Argand diagram	30 min
Modulus and argument	25 min
Homework	5 min

LEARNING OBJECTIVES

- Add, subtract, multiply and divide complex numbers in the form $a + bi$.
- Use the conjugate to divide, and to state the roots of a real quadratic.
- Plot a complex number on the Argand diagram.
- Find the modulus and argument, and convert to polar form.

TEXTBOOK

National	pp. 411–428
Cambridge	Pure Mathematics 2 & 3, pp. 246–262
Workbook	Exercise 11A–11D

01

Explanation

The new number

The equation $x^2 + 1 = 0$ has no real solution. Define one:

$i^2 = -1$

Then every number of the form $z = a + bi$, with a and b real, is a **complex number**: $a = \operatorname{Re}(z)$ is the real part, $b = \operatorname{Im}(z)$ the imaginary part. Note that $\operatorname{Im}(z)$ is the real number b , not bi .

Z	$RE(Z)$	$IM(Z)$	KIND
$3 + 4i$	3	4	complex
$-2i$	0	-2	purely imaginary
7	7	0	real

POWERS OF i CYCLE WITH PERIOD FOUR

$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$

So i^n depends only on n modulo 4: $i^{2026} = i^2 = -1$, because $2026 = 4 \times 506 + 2$.

TWO COMPLEX NUMBERS ARE EQUAL ONLY IF BOTH PARTS MATCH

$a + bi = c + di$ means $a = c$ **and** $b = d$. This turns one complex equation into two real ones — the standard technique for finding an unknown complex number.

Arithmetic

OPERATION	RULE	EXAMPLE
add	parts separately	$(3+4i) + (1-2i) = 4 + 2i$
subtract	parts separately	$(3+4i) - (1-2i) = 2 + 6i$
multiply	expand, then $i^2 = -1$	$(3+4i)(1-2i) = 11 - 2i$
square	as a binomial	$(3+4i)^2 = -7 + 24i$

The multiplication in full: $(3+4i)(1-2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$.

Nothing is new except the last step, where $-8i^2$ becomes $+8$.

EVERY QUADRATIC NOW HAS TWO ROOTS

When $D < 0$ the formula still works: $\sqrt{-16} = 4i$, so $x^2 - 2x + 5 = 0$ gives $x = \frac{2 \pm 4i}{2} = 1 \pm 2i$. Real quadratics with $D < 0$ have **conjugate** roots — always a pair.

The conjugate and division

$$z = a + bi \Rightarrow \bar{z} = a - bi, z \bar{z} = a^2 + b^2 \text{ — always real and non-negative}$$

That last identity is the whole reason division works. To divide, multiply top and bottom by the conjugate of the bottom:

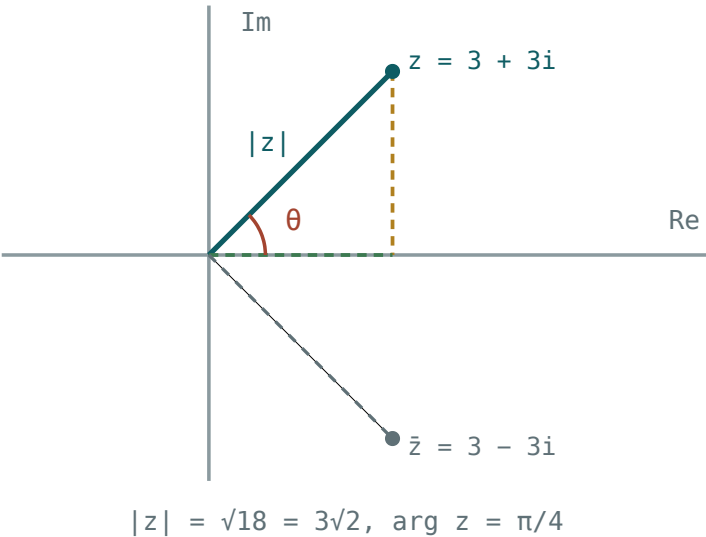
$$\frac{3 + 4i}{1 - 2i} = \frac{(3 + 4i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{-5 + 10i}{5} = -1 + 2i$$

PROPERTY	STATEMENT
sum	$z + \bar{z} = 2\text{Re}(z)$
difference	$z - \bar{z} = 2i \cdot \text{Im}(z)$
product	$z \bar{z} = z ^2$
of a sum	$(z + w)^{-} = \bar{z} + \bar{w}$
of a product	$(zw)^{-} = \bar{z} \bar{w}$

The last two say that conjugation respects arithmetic — which is why the non-real roots of a **real** polynomial always come in conjugate pairs.

The Argand diagram

Plot $a + bi$ at the point (a, b) : the horizontal axis is the real one, the vertical the imaginary. The plane of complex numbers is the **Argand diagram**.



z as a point, its modulus as a distance, its argument as an angle — and $\bar{z}$ as the reflection in the real axis.

OPERATION	ON THE DIAGRAM
addition	vector addition — the parallelogram rule
conjugation	reflection in the real axis
multiplying by -1	rotation by 180°
multiplying by i	rotation by 90° anticlockwise

MULTIPLYING BY i IS A QUARTER TURN

$i(a + bi) = ai + bi^2 = -b + ai$: the point (a, b) moves to $(-b, a)$, which is exactly a 90° rotation about the origin. This single fact makes complex numbers a tool for geometry.

Modulus and argument

$$|z| = \sqrt{a^2 + b^2}, \arg z = \theta \text{ with } \tan \theta = \frac{b}{a}$$

The modulus is the distance from the origin; the argument is the angle from the positive real axis, measured anticlockwise and normally taken in $(-\pi, \pi]$.

THE QUADRANT DECIDES THE ARGUMENT, NOT THE CALCULATOR

For $z = -1 + i$, $\tan \theta = -1$ and a calculator returns -45° . But the point is in the **second** quadrant, so $\arg z = 135^\circ = \frac{3\pi}{4}$. Always plot first.

$$z = r(\cos \theta + i \sin \theta) \text{ — the polar form}$$

Z	$ Z $	$ARG\ Z$	POLAR
$1 + i$	$\sqrt{2}$	$\frac{\pi}{4}$	$\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$
$-1 + i$	$\sqrt{2}$	$\frac{3\pi}{4}$	$\sqrt{2}(\cos \frac{3\pi}{4} + \dots)$
$-2i$	2	$-\frac{\pi}{2}$	$2(\cos(-\frac{\pi}{2}) + \dots)$
3	3	0	$3(\cos 0 + i \sin 0)$

WHY POLAR FORM IS WORTH THE TROUBLE

$$|zw| = |z||w| \quad arg(zw) = arg\ z + arg\ w$$

Multiplication **multiplies** the moduli and **adds** the arguments. A product that takes four lines in Cartesian form takes one in polar form — and powers become trivial.

02 Worked examples

EXAMPLE 1 Simplify $(3 + 4i)(1 - 2i)$ and $\frac{3 + 4i}{1 - 2i}$.

$$(1) \quad (3+4i)(1-2i) = 3 - 6i + 4i - 8i^2$$

$$(2) \quad = 3 - 2i + 8 = 11 - 2i$$

$$(3) \quad \frac{(3+4i)(1+2i)}{(1-2i)(1+2i)} = \frac{3 + 6i + 4i - 8}{5}$$

Multiply by the conjugate.

$$(4) \quad = \frac{-5 + 10i}{5} = -1 + 2i$$

ANSWER

$11 - 2i$ and $-1 + 2i$

EXAMPLE 2 Solve $x^2 - 2x + 5 = 0$.

$$(1) \quad D = 4 - 20 = -16$$

$$(2) \quad \sqrt{-16} = 4i$$

$$(3) \quad x = \frac{2 \pm 4i}{2}$$

$$(4) \quad x = 1 \pm 2i$$

A conjugate pair.

ANSWER

$$x = 1 + 2i \text{ and } x = 1 - 2i$$

EXAMPLE 3 Find $|z|$ and $\arg z$ for $z = -1 + i$.

$$(1) \quad |z| = \sqrt{1 + 1} = \sqrt{2}$$

$$(2) \quad \tan \theta = -1$$

The calculator says -45° .

$$(3) \quad \text{The point is in the second quadrant.}$$

$$(4) \quad \arg z = \frac{3\pi}{4}$$

ANSWER

$$|z| = \sqrt{2}, \arg z = \frac{3\pi}{4}$$

EXAMPLE 4 Find real x and y with $(x + yi)(2 - i) = 5 + 5i$.

① $2x - xi + 2yi - yi^2 = (2x + y) + (2y - x)i$

② $2x + y = 5$ and $2y - x = 5$

Equate parts.

③ $x = 1, y = 3$

④ Check: $(1 + 3i)(2 - i) = 2 - i + 6i + 3 = 5 + 5i$.

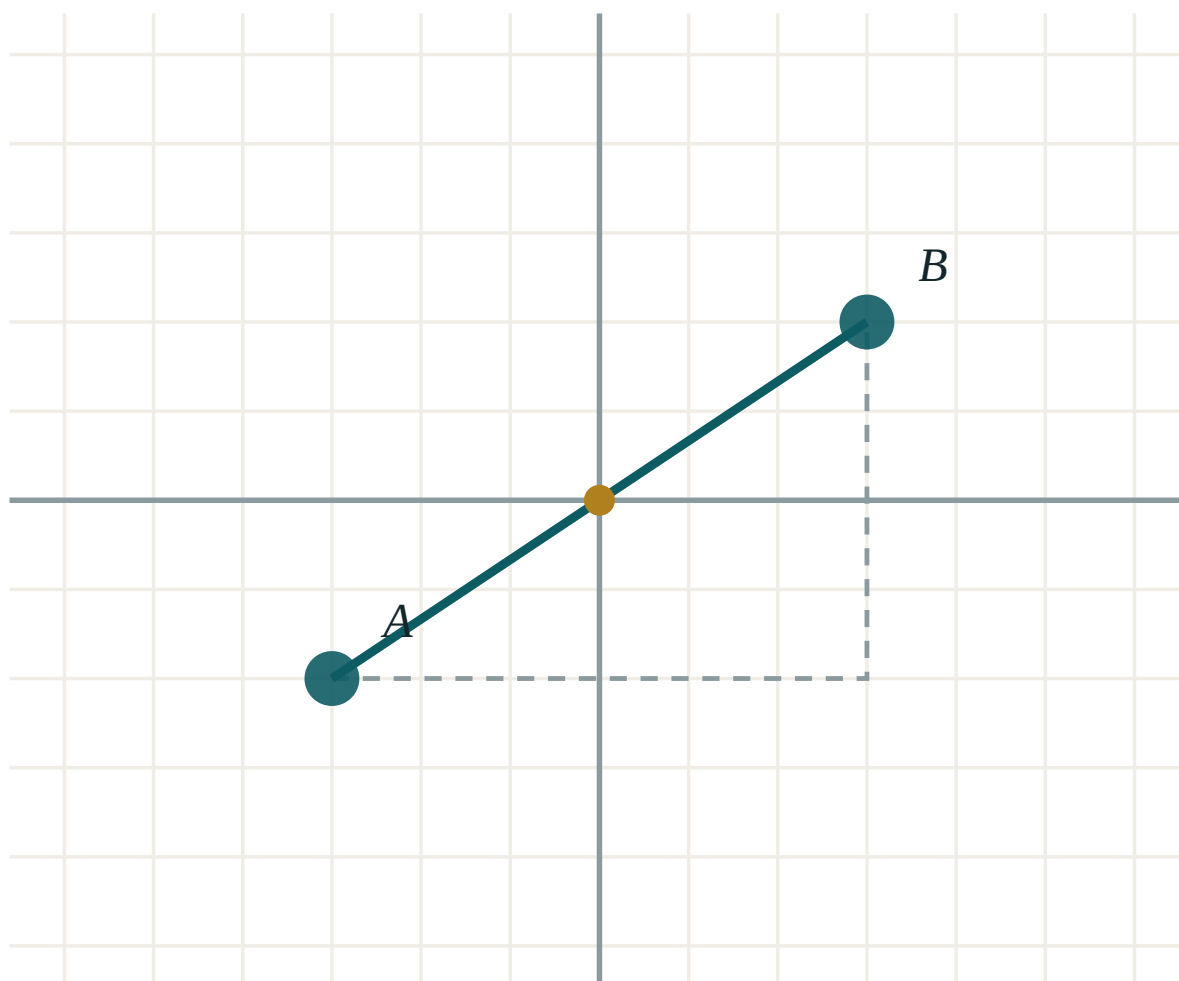
ANSWER

$x = 1, y = 3$

03 Interactive model

Draw the Argand diagram once and do every example on it — the geometry makes the algebra memorable.

The complex plane



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Complex number	Kompleks son	Комплексное число
Imaginary unit	Mavhum birlik	Мнимая единица
Real part	Haqiqiy qism	Действительная часть
Imaginary part	Mavhum qism	Мнимая часть
Conjugate	Qo'shma son	Сопряжённое число
Argand diagram	Argan diagrammasi	Диаграмма Аргана
Modulus	Modul	Модуль
Argument	Argument	Аргумент
Polar form	Trigonometrik shakl	Тригонометрическая форма
Complex plane	Kompleks tekislik	Комплексная плоскость
Purely imaginary	Sof mavhum	Чисто мнимое
Equating parts	Qismlarni tenglashtirish	Приравнивание частей

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** i^2 equals: i^{2026} equals: $Im(3 + 4i)$ is:

To divide, multiply by:

the reciprocal

the conjugate of the denominator

i

the modulus

$z \bar{z}$ equals:

$|z|$

$|z|^2$

$2\text{Re}(z)$

0

Conjugation on the Argand diagram is:

a rotation

reflection in the real axis

an enlargement

a translation

Multiplying by i is:

90° anticlockwise

180°

a reflection

a doubling

$\arg(-1 + i)$ is:

$$-\frac{\pi}{4}$$

$$\frac{3\pi}{4}$$

$$\frac{\pi}{4}$$

$$-\frac{3\pi}{4}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(3 + 4i) + (1 - 2i)$

ANSWER $4 + 2i$

2. $(3 + 4i) - (1 - 2i)$

ANSWER $2 + 6i$

3. i^3

ANSWER $-i$

4. i^4

ANSWER 1

5. $|3 + 4i|$

ANSWER 5

6. Conjugate of $2 - 5i$

ANSWER $2 + 5i$

7. $Re(7)$ and $Im(7)$

ANSWER $7, 0$

Medium · the standard question

7 PROBLEMS

1. $(3 + 4i)(1 - 2i)$

ANSWER $11 - 2i$

2. $(3 + 4i)^2$

ANSWER $-7 + 24i$

3. $\frac{3 + 4i}{1 - 2i}$

ANSWER $-1 + 2i$

4. $\frac{1}{i}$

ANSWER $-i$

5. Solve $x^2 - 2x + 5 = 0$

ANSWER $1 \pm 2i$

6. $|-1 + i|$ and $\arg(-1 + i)$

ANSWER $\sqrt{2}, \frac{3\pi}{4}$

7. i^{2026}

ANSWER -1 **Hard · stretch and reason**

7 PROBLEMS

1. Find real x, y with $(x + yi)(2 - i) = 5 + 5i$

ANSWER $x = 1, y = 3$

2. Solve $z^2 = 3 + 4i$

ANSWER $z = \pm(2 + i)$

3. Write $1 + \sqrt{3}i$ in polar form

ANSWER $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

4. Show that $(1 + i)^8 = 16$

ANSWER $|z| = \sqrt{2}, \arg = \frac{\pi}{4}, \text{ so } 16(\cos 2\pi + i \sin 2\pi)$

5. One root of $x^3 - 5x^2 + 17x - 13 = 0$ is 1: find the others

ANSWER $2 \pm 3i$

6. Prove that $z + \bar{z}$ is always real and $z - \bar{z}$ always purely imaginary

ANSWER $2a$ and $2bi$

7. Simplify $\frac{(1 + i)^3}{1 - i}$

ANSWER -2

07 Homework

Homework — 5 tasks

Plot every answer on an Argand diagram; the picture catches quadrant errors at once.

1. Simplify $(2 - 3i)(4 + i)$ and $\frac{2 - 3i}{4 + i}$.
2. Solve $x^2 + 4x + 13 = 0$ and plot both roots on an Argand diagram.
3. Find $|z|$ and $\arg z$ for $z = -3 - 3i$, and write z in polar form.

4. Find real x and y such that $(x + yi)(1 + 2i) = 4 - 3i$.
5. Show that i^n takes only four values, and evaluate i^{100} and i^{2027} .